Compiler Construction

Lab 4

Submitted By:

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BSCS-6

Task 1:

Lab 4(Mannoor).cpp

```cpp
#include<iostream>
using namespace std;

int main(){

string s;
cin>>s;

int state =0;
for(char c:s){
    if(state==0){
        if(c=='0') state =1;
        else state =0;}
    else
    {
        state=1;
    }

    if(state==1)cout<<"Accepted\n";
    else cout<<"Rejected";
}
}
```

C:\Users\Riphah\

```
0111
Accepted
Accepted
Accepted
Accepted

--------------
Process exited
Press any key t
```

Task 2:

CC Lab 4(Mahnoor).cpp Task 2.cpp

```cpp
#include<iostream>
using namespace std;

int main(){
    string s;
    cin>>s;

    int state =0;

    for(char c:s){
        if(state ==0){
            if (c=='1')state =1;
            else state =3;
        }

        else if(state ==1){
            if(c=='0') state =2;
            else state =1;
        }
        else { state =3; //this is dead state
        }
    }

    if(state ==2)
    cout<<"Accepted";
    else
    cout<<"Rejected";
}
```

C:\Users\Riphah\Desktop\Lab 4\Task 2.exe

```
10
Accepted
--------------------------------
Process exited after 2.021 second
Press any key to continue . . .
```